

* Information is a fⁿ of prob: it is the Entropy associated w/ the prob. distribution.

• Two slit particle: $P = P_1 + P_2$

• In QM each slit is ψ_1 or ψ_2

and so

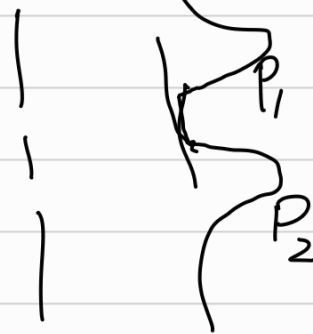
$$P = |\psi_1 + \psi_2|^2$$

$$\psi_1 + \psi_2 = \psi$$

Thus,

$$P = |\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + \psi_1 \psi_2^* + \psi_1^* \psi_2$$

$$P_1 = |\psi_1|^2, P_2 = |\psi_2|^2$$



~~$$|\vec{x} - \vec{y}|^2 = |\vec{x}|^2 + |\vec{y}|^2 + 2|\vec{x}||\vec{y}|\cos(\theta)$$~~

$$= (\psi_1 + \psi_2) \cdot (\bar{\psi}_1 + \bar{\psi}_2)$$

$$= |\psi_1|^2 + |\psi_2|^2 + \psi_2 \cdot \psi_1^* + \psi_1 \cdot \psi_2^*$$

$$= |\psi_1|^2 + |\psi_2|^2 + |\psi_2||\psi_1|\cos(\theta) + |\psi_1||\psi_2|\cos(\theta)$$

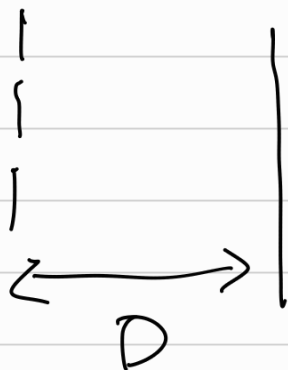
$$= |\psi_1|^2 + |\psi_2|^2 + 2|\psi_1||\psi_2|\cos(\theta)$$

$$= |\psi_1|^2 + |\psi_2|^2 + 2\sqrt{P_1 P_2} \cos(\theta) \quad (\text{Wave solution})$$

• The wave solution is one in which we do not monitor the slits.

$$|\psi_1\rangle + |\psi_2\rangle = |\psi\rangle$$

* The phase θ : depends on the distance D to the wall from the slit:



Conditional Probabilities:

• Event: A

Event: A
Outcomes: $\{a_i\}$

→ Prob. that $a_i = A$ occurs is given by $P(a_i)$
 $\Rightarrow \sum_i P(a_i) = 1$

* We then add an event $B: \{b_j\}$

Then $P(a_i)$ and $P(b_j)$ are not the complete set of info.

→ Must include the JOINT Prob.

That is, each event can occur separately however there is further information which occurs when we measure both event simultaneously.

Joint Prob: $\{P(a_i, b_j)\}$

$P(a_i, b_j)$

← Comma means AND ($\wedge$)

$P(a_i \wedge b_j)$

* If $P(a_i)$ and $P(b_j)$ are independent then $P(a_i \wedge b_j) = P(a_i)P(b_j)$

* Note we can obtain single event prob. by summing over an event:

$$P(a_i) = \sum_j P(a_i \wedge b_j)$$

$$P(b_j) = \sum_i P(a_i \wedge b_j)$$

* What if we know the prob. of A ?

$A = a_0$: the Quantities are Conditional:

$$\{P(b_j | a_0)\}$$

$P(b_j | a_0)$: Prob of b_j given that $A = a_0$

- Obviously, this means that $P(b_j | a_0)$ is proportional to $P(b_j \wedge a_0)$

$$P(b_j | a_0) = K(a_0) P(b_j \wedge a_0)$$

$$\sum_j P(b_j | a_0) = 1$$
$$\therefore 1 = K(a_0) \sum_j P(b_j \wedge a_0)$$

$$\Rightarrow K(a_0) = \left(\sum_j P(b_j \wedge a_0) \right)^{-1}$$

Recall $P(a_k) = \sum_i P(b_i \wedge a_k)$

$$\therefore K(a_0) = P(a_0)^{-1}$$

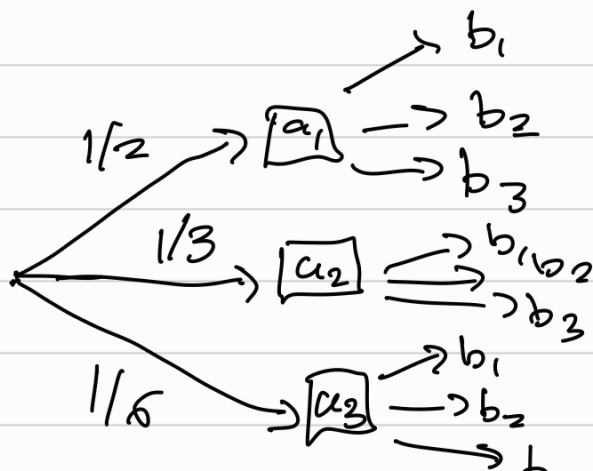
$$\Rightarrow P(a_0 \wedge b_j) = P(a_0) P(b_j | a_0)$$

In general: $P(a_i, b_j) = P(a_i) P(b_j | a_i)$

could ask $P(a_i | b_j) P(b_j) = P(a_i \wedge b_j)$

Ex: Prob tree: $A = a_1 \vee a_2 \vee a_3$

1a



$$P(b_j | a_i)$$

$$\begin{aligned} P(a_1) &\equiv \frac{1}{2} \\ P(a_2) &\equiv \frac{1}{3} \\ P(a_3) &\equiv \frac{1}{6} \end{aligned}$$

Thus: $P(a_i) P(b_j | a_i) = P(a_i \wedge b_j)$

ex: $\frac{1}{2} P(b_j | a_1) = P(a_1 \wedge b_j)$

$$P(b_j | a_1) = 2 P(a_1 \wedge b_j)$$

Given in text:

Since

$$P(a_i) P(b_j | a_i) = P(a_i \wedge b_j)$$

$$P(b_j) P(a_i | b_j) = P(b_j \wedge a_i)$$

∴ Baye's Rule:

$$P(b_j | a_i) = \frac{P(b_j) P(a_i | b_j)}{P(a_i)}$$



Event: Arriving on time:

A_{Long} , A_{Short}

B_0 : on time, B_L : Late

$$\Leftrightarrow P(B_0 | A_S) = 1$$

$$P(B_0 | A_2) = 3/4$$

$$P(A_L) = 2/4$$

$$P(A_S) = 3/4 \quad \leftarrow \text{from above}$$

$$P(B_i | A_j) = \frac{P(B_i) P(A_j | B_i)}{P(A_i)}$$

$$P(B_0 | A_j) = \{P(B_0) P(A_j | B_0)\} / P(A_i)$$

$$(i) A_S: P(B_0 | A_S) = 1 = \frac{P(B_0) P(A_S | B_0)}{P(A_S)}$$

$$\frac{3}{4} = P(B_0) P(A_S | B_0)$$

$$(ii) A_L: P(B_0 | A_L) = \frac{3}{4} = \frac{P(B_0) P(A_L | B_0)}{P(A_L)}$$

$$4 P(A_S | B_0) = 4 P(A_L | B_0)$$

$$P(A_S | B_0) = P(A_L | B_0)$$

$$\sum_i P(a_i | b_j) = 1$$

Recall!

$$P(b_0) = \sum_j P(b_0 \cap A_j)$$

$$= \sum_j P(A_j) P(b_0 | A_j)$$

$$\begin{aligned} \therefore P(B_0) &= P(A_S) P(b_0 | A_S) + P(A_L) P(b_0 | A_L) \\ &= \left(\frac{3}{4}\right) \left(1\right) + \left(\frac{1}{4}\right) \left(\frac{3}{4}\right) \\ &= \frac{3}{4} + \frac{3}{16} = \frac{15}{16} \end{aligned}$$

$$P(B_0) = \frac{15}{16}$$

$$P(B_L) = 1/16$$

$$P(A_L | B_0) = \frac{P(A_L) P(B_0 | A_L)}{P(B_0)}$$

$$= (1/4)(3/4) / (13/16)$$

$$= 3/13 = 1/5 \quad \checkmark$$

$$(1/5)$$

Proof: (Claim $\sum_i P(A_i | B) = 1$)

$$P(A_i) = \sum_k P(A_i \cap B_k)$$

$$= \sum_k P(A_i | B_k) P(B_k)$$

$$\sum_i P(A_i) = 1 = \sum_i \sum_k P(A_i | B_k) P(B_k)$$

$$= \sum_k P(B_k) \sum_i P(A_i | B_k)$$

$$1 = \sum_k P(B_k) f(B_k)$$

But $\sum_k P(B_k) = 1$ so

$$\sum_i P(A_i | B_k) = 1 \quad \checkmark$$

$$P(A | B_j) \propto P(A \cap B_j) = g(A) \sum_j P(A \cap B_j)$$

$$\sum_j P(B_j | A) = g(A) P(A)$$

Must be unity

$\therefore$ By what I had:

$$4 P(1) + P(1) = 1$$

$$P(1) = 1/5$$

$$\text{and } 1 - P(1) = 4/5.$$

• This would have worked as well.

* How an event affects one another: Fisher's rule

$\rightarrow$ If we discover that $B = b_j$, then

$$P(a_i | b_j) = l(a_i | b_j) P(a_i)$$

$l(a_i | b_j) :=$ likelihood of a_i given b_j .

* Three events: $P(a_i, b_j, c_k) = P(a_i | b_j, c_k) P(b_j, c_k)$
 $P(b_j, c_k) = P(c_k) P(b_j | c_k)$

$\therefore$

$$P(a_i, b_j, c_k) = P(a_i | b_j, c_k) P(c_k) P(b_j | c_k)$$

• We also have,

$$P(a_i, b_j, c_k) = P(a_i, b_j | c_k) P(c_k)$$

$$P(a_i | b_j, c_k) P(c_k) P(b_j | c_k) = P(a_i, b_j | c_k) P(c_k)$$

$$\therefore \boxed{P(a_i, b_j | c_k) = P(a_i | b_j, c_k) P(b_j | c_k)}$$

$$P(a_i, b_j, c_k) = P(a_i | b_j, c_k) P(b_j, c_k) P(c_k)$$

Bayes Rule:

$$P(a_i | b_j, c_k) = \frac{P(b_j, c_k | a_i) P(a_i)}{P(b_j, c_k)}$$

Entropy & Information:

- Define information as the entropy of the associated probability distribution for an event.

App. A: The Eq. of Info & Entropy

Assumption: Suppose that an event E has n possible outcomes: i.e. n sided dice.
 $\{p_1, \dots, p_n\}$ = set of outcomes

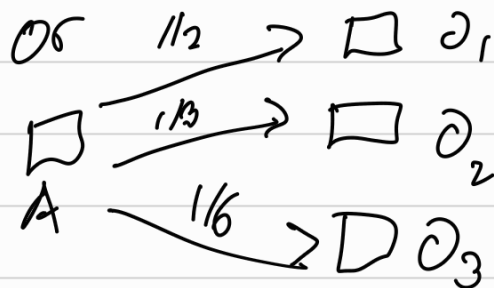
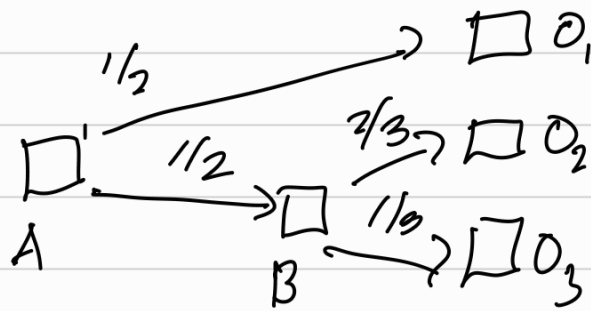
- We would like to know the properties of H
 $H = H(p_1, \dots, p_n)$
 which reflects the amount of choice involved in the selection of the event, or, the degrees of uncertainty as to the outcome.

Require:

- H should be continuous
- If $p_1 = p_2 = \dots = p_n$ then $p_i = \frac{1}{n}$, then H should be monotonically increasing $\frac{1}{n}$ of n .
- i.e. If $p_i \rightarrow \begin{matrix} p_{i1} \\ p_{i2} \\ \vdots \end{matrix}$ is broken down into others

choices then $P_{i,k}$ H should be the prob. weighted sum of the individual values of H .

ex:-



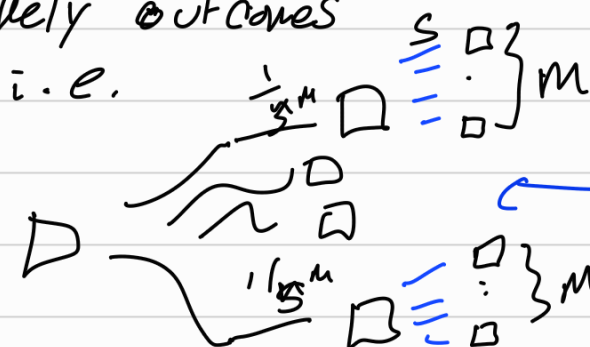
each has the same outcome. Thus we should expect

$$H(1/2, 1/3, 1/6) = H(1/2, 1/2) + \frac{1}{2} H(2/3, 1/3)$$

The factor $1/2$ comes from the fact that $H(2/3, 1/3)$ occurs half the time.

Suppose $p_1 = p_2 = \dots = p_n$ such that $p_i = \frac{1}{n} \forall i$
 $H(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}) \equiv A(n)$

Then for an event w s^m equal prob. into a series of m events, each w/ s equally likely outcomes



each - has s out
 come of s

$$H(\frac{1}{n}, \dots, \frac{1}{n}) = \frac{1}{n} H(s, \dots, s) + \frac{1}{n} H(s, \dots, s)$$

$$\sum_{i=1}^m S = 1 \Leftrightarrow S(1+2+\dots+m) = 1$$

$$\frac{m(m+1)}{2} \stackrel{!}{=} \sum_{i=1}^m 1$$

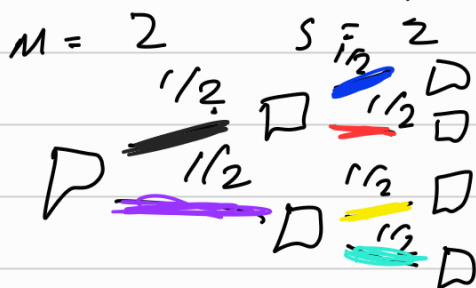
$$m=2: \frac{3}{2}$$

$$m=3: \frac{3 \cdot 4}{2} = 6 \stackrel{!}{=} 1+2+3 = 6$$

$$\left[m(m+1)/2 \right] S = 1$$

$$A(S^m) = H\left(\frac{1}{S^m}, \frac{1}{S^m}, \dots, \frac{1}{S^m}\right)$$

ex: m : even, S : outcomes



$$S^m = 4$$

Possible outcomes:

(i) $\rightarrow$ $\rightarrow$

$\rightarrow$ $\rightarrow$

(ii) $\rightarrow$ $\rightarrow$

$\rightarrow$ $\rightarrow$

(iii) $\rightarrow$ $\rightarrow$

$\rightarrow$ $\rightarrow$

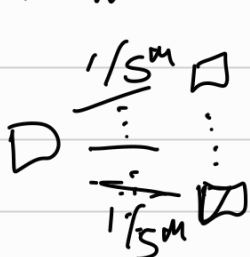
(iv) $\rightarrow$ $\rightarrow$

$\rightarrow$ $\rightarrow$

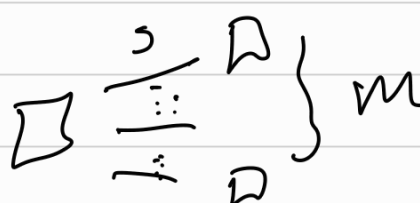
i.e. 4

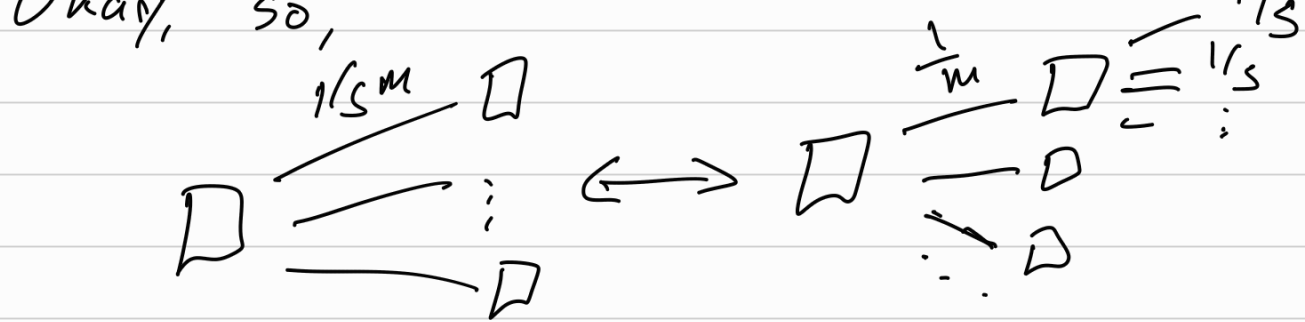
$$H\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} H\left(\frac{1}{2}, \frac{1}{2}\right) + \frac{1}{2} H\left(\frac{1}{2}, \frac{1}{2}\right)$$

? ... No.



$\leftrightarrow$





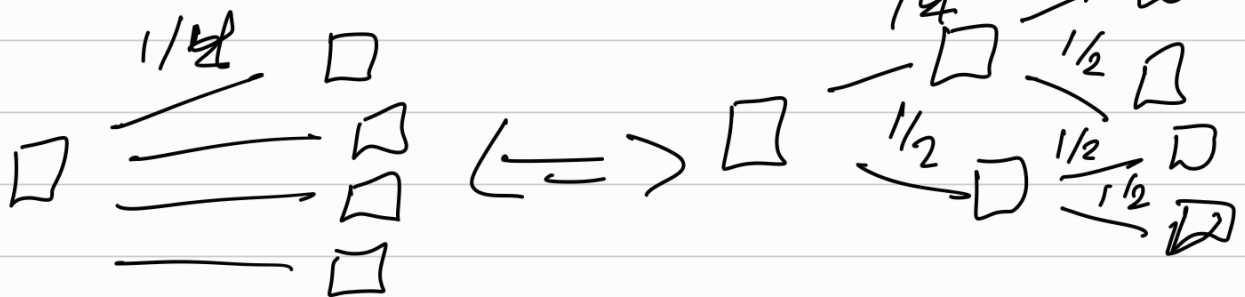
$$H(\underbrace{\frac{1}{s^m}, \dots, \frac{1}{s^m}}_{s^m}) = H(\underbrace{\frac{1}{s^m}, \dots, \frac{1}{s^m}}_m) + \frac{1}{s^m} H(\underbrace{\frac{1}{s}, \dots, \frac{1}{s}}_s)$$

$$H(\frac{1}{s^m}, \dots, \frac{1}{s^m})^{(s^m)} = H^{(m)}(\frac{1}{m}, \dots, \frac{1}{m}) + \frac{1}{m} H(\frac{1}{s}, \dots, \frac{1}{s})^{(s)} + \dots$$

$m=1, S=2$;

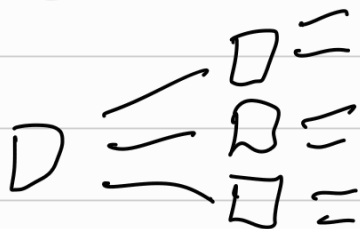
Diagram showing a root node branching into two children, each labeled $1/s^m$. Same.

$m=2, S=2$



$M=3, S=2$

$2^3 = 16$



$$H(\frac{1}{s^m}, \dots, \frac{1}{s^m}) = H^{(m)}(\frac{1}{m}, \dots, \frac{1}{m}) + H(\frac{1}{s}, \dots, \frac{1}{s})$$

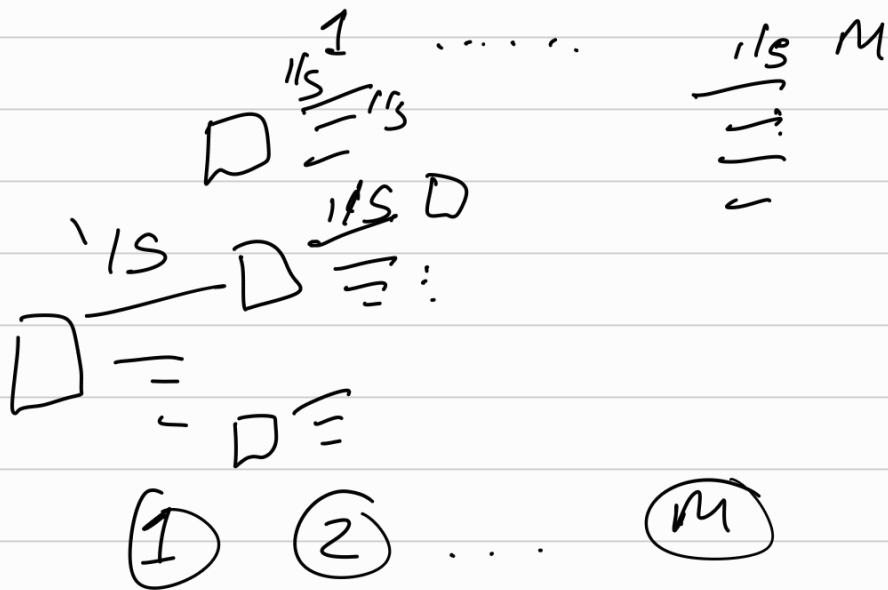
$$H(s^m) = H(m) + H(s)$$

Alt. an event is one step to the Right!

$\therefore$ 0. $\textcircled{1}$ $\textcircled{2}$ $\textcircled{3}$ etc....

$\square \xrightarrow{\quad} \square \xrightarrow{\quad} \square \xrightarrow{\quad} \square$
 $\square \xrightarrow{\quad} \square \xrightarrow{\quad} \square \xrightarrow{\quad} \square$

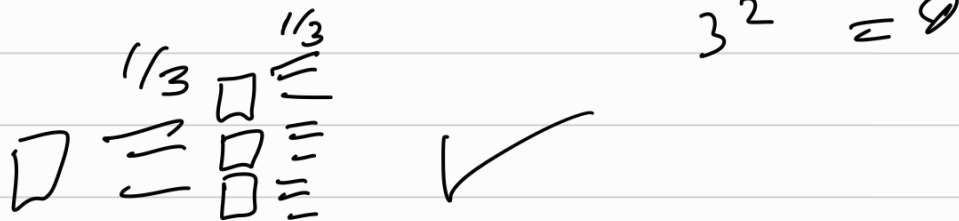
I see... okay, so then



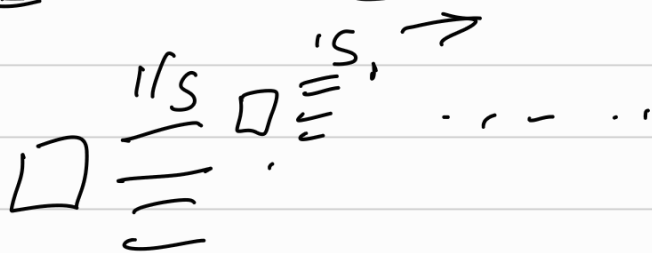
$$M = 2, S = 2$$



$$M = 2, S = 3$$



I Seeee...



$$H\left(\frac{1}{s^m}, \dots, \frac{1}{s^m}\right) = H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \left\{ \left[\frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) \right] + \dots \right\}$$

$$\begin{aligned} \frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) &= \frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \frac{1}{s^2} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) \\ &= \frac{1}{s} \left[H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) \right] \\ &= \frac{1}{s} \left[H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \frac{1}{s} \left[H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) \right] \right] \\ &= \frac{1}{s} \left[H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \frac{1}{s^2} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) \right] \end{aligned}$$

①

②

$$H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \dots + \frac{1}{s} H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \dots$$

$$= H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + H\left(\frac{1}{s}, \dots, \frac{1}{s}\right) + \dots + H\left(\frac{1}{s}, \dots, \frac{1}{s}\right)$$

① ② m

$$\therefore = m H\left(\frac{1}{s}, \dots, \frac{1}{s}\right)$$

out comes has s^m :

$$\begin{array}{c} \square \\ \square \\ \square \\ \square \\ \square \end{array}$$

$$\left. \begin{array}{c} \square \\ \square \\ \square \\ \square \\ \square \end{array} \right\} s^m$$

$$H\left(\frac{1}{s^m}, \dots, \frac{1}{s^m}\right) = m H\left(\frac{1}{s}, \dots, \frac{1}{s}\right)$$

$$A(x) := H\left(\frac{1}{x}, \dots, \frac{1}{x}\right)$$

$$A(s^m) = m A(s)$$

Now, consider set up w/ n steps and t outcomes: $A(t^n) = nA(t)$

Let n be large s.t. $n \gg 1$

$$s^m \leq t^n \leq s^{m+1}$$

$$m \log(s) \leq n \log(t) \leq (m+1) \log(s)$$

$$m/n \leq \frac{\log(t)}{\log(s)} \leq \frac{(m+1)}{n}$$

$$0 \leq \frac{\log(t)}{\log(s)} \leq \frac{1}{n}$$

Since $n \gg 1$: $1/n \equiv \epsilon$

$$0 \leq \frac{\log(t)}{\log(s)} - \frac{m}{n} \leq \epsilon \Rightarrow \left| \frac{m}{n} - \frac{\log(t)}{\log(s)} \right| < \epsilon$$

By the monotonicity of H

$$A(s^m) \leq A(t^n) \leq A(s^{m+1})$$

$$mA(s) \leq nA(t) \leq (m+1)A(s)$$

$$\left| \frac{m}{n} - \frac{A(t)}{A(s)} \right| < \epsilon$$

$$\|f\| \leq U \cdot \|f\| + \|f\|$$

$\therefore$ Adding the above, we find:

$$\left| \frac{m}{n} - \frac{A(t)}{A(s)} \right| + \left| \frac{m}{n} - \frac{\log(t)}{\log(s)} \right| < 2\epsilon$$

$$\left| \frac{m}{n} - \frac{\log(t)}{\log(s)} \right| + \left| \frac{m}{n} - \frac{A(t)}{A(s)} \right| < 2\epsilon$$

$$\forall \left| \frac{1}{n} - \frac{1}{\log(s)} - \left(\frac{1}{n} - \frac{1}{A(s)} \right) \right| < 2 \in$$

$$\therefore \left| \frac{A(t)}{A(s)} - \frac{\log(t)}{\log(s)} \right| < 2 \in$$

$$\therefore A(t) = 1 < \log(2t)$$

For probabilities $p_i = n_i / \sum_j n_j$ which are not equal

$$\begin{aligned} 1 &= \sum_i p_i = \frac{n_1}{S} \\ &\vdots \\ n &= \sum_i p_n = \frac{n_n}{S} \end{aligned}$$

$$\bullet \sum_i n_i \approx S^m$$

$$\hookrightarrow \begin{aligned} &\frac{p_1}{p_i} \approx \frac{1/n_i}{1/n_i} \approx 1 \\ &\vdots \\ &\frac{p_n}{p_i} \approx \frac{1/n_i}{1/n_i} \approx 1 \end{aligned} \left. \vphantom{\frac{p_1}{p_i}} \right\} n_i$$

$$\Rightarrow \sum_i n_i = \# \text{ of states}$$

$$A(\sum_i n_i) = \text{"RHS"}$$

$$A(\sum_i p_i) = H(p_1, \dots, p_n) + \sum_i p_i A(n_i)$$

$$K \log(\sum_i n_i) = H(p_1, \dots, p_n) + \sum_i p_i K \log(n_i)$$

$$\begin{aligned} \therefore H(p_1, \dots, p_n) &= K \left[\log(\sum_i n_i) - \sum_i p_i \log(n_i) \right] \\ &= K \left[\sum_i p_i \log(\sum_k n_k) - \sum_i p_i \log(n_i) \right] \\ &= K \sum_i p_i \log \left(\frac{\sum_k n_k}{n_i} \right) \end{aligned}$$

$$= K \sum_i p_i \log \left(\frac{n_i}{p_i} \right) \quad p_i = \frac{n_i}{S}$$

$$H(p_1, \dots, p_n) = -K \sum_i p_i \log(p_i)$$

Back to Entropy & Information:

A: event {outcomes} $\begin{matrix} \text{A} \\ \hline a_1 \\ \hline \vdots \\ \hline a_n \end{matrix}$

$h(p(a_i)) =$ "info we obtain in learning $A = a_i$ "

$$P(a_i, b_j) = P(a_i) P(b_j)$$

$$\log_x(a) = \frac{\log_b(a)}{\log_b(x)}$$

Proof:

$$\begin{aligned} x^{\log_x(a)} &= a \\ \log_b(x^{\log_x(a)}) &= \log_b(a) \\ \log_x(a) \log_b(x) &= \log_b(a) \end{aligned}$$

$$\log \equiv \log_2$$

* Binary system: {0, 1}
then

let 0 have prob p .
then 1 has $1-p$

$$H = - \sum_i p_i \log(p_i)$$

$$\begin{aligned} H &= -p \log(p) - (1-p) \log(1-p) \\ \frac{dH}{dp} &= -\log(p) - 1 + \log(1-p) + 1 \stackrel{!}{=} 0 \\ \log\left(\frac{1}{p} - 1\right) &= 0 \\ \frac{1}{p} - 1 &= 1 \Leftrightarrow \boxed{p_{\max} = \frac{1}{2}} \end{aligned}$$

Convex f'' : $f(x)$: on (a, b) , $\forall x_i, x_j, \lambda \in [0, 1]$

$$f[\lambda x_i + (1-\lambda)x_j] \leq \lambda f(x_i) + (1-\lambda)f(x_j)$$

$$\therefore f\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i f(x_i)$$

• f'' is Concave if $-f(x)$ is convex

* For $a_1 = \dots = a_n$, $p_i = \frac{1}{n}$

$$H(A) = - \sum_i \frac{1}{n} \log\left(\frac{1}{n}\right) = \log(n)$$

• Information increases the more we know the
 $P(a_i) = \sum_j \lambda_{ij} P(a_j)$ $\sum_i \lambda_{ij} = 1 = \sum_j \lambda_{ij}$

$$\lambda_{ij} \geq 0$$

* Events A & B : $H(A, B) = - \sum_{ij} P(a_i, b_j) \log(P(a_i, b_j))$

$$\text{Recall, } P(a_i) = \sum_j P(a_i, b_j)$$

$$H(A) = - \sum_{ij} P(a_i, b_j) \log\left[\sum_k P(a_i, b_k)\right]$$
$$H(B) = - \sum_{ij} P(a_i, b_j) \log\left[\sum_k P(a_k, b_j)\right]$$

$$P(a_i, b_j) = P(a_i | b_j) P(b_j)$$

(i) Subadditivity:

$$H(X, Y) \leq H(X) + H(Y)$$

• Equality iff $X \cap Y = \emptyset$

Proof: $\log(x) \leq \frac{(x-1)}{\ln(2)}$

→ subproof:

$$\ln(x) \leq 1 - x$$

$$\frac{P(a_i)P(b_j)}{1}$$

$$\sum_{i,j} P(a_i, b_j) \log \left(\frac{P(a_i) P(b_j)}{P(a_i, b_j)} \right) \leq \sum_{i,j} P(a_i, b_j) \frac{P(a_i) P(b_j) - 1}{\ln(2)}$$

$$= \frac{1}{\ln(2)} \left[\sum_{i,j} P(a_i, b_j) \frac{P(a_i) P(b_j)}{P(a_i, b_j)} - 1 \right] = 0$$

$$\therefore \sum_{i,j} P(a_i, b_j) \log \left(\frac{P(a_i) P(b_j)}{P(a_i, b_j)} \right) \leq 0$$

$$\sum_{i,j} P(a_i, b_j) \log(P(a_i)) + \sum_{i,j} P(a_i, b_j) \log(P(b_j))$$

$$\leq \sum_{i,j} P(a_i, b_j) \log(P(a_i, b_j))$$

→ Multiply by (-1)

$$H(A) + H(B) \geq H(A, B)$$

$$* H(A) + H(B) - H(A, B) \equiv H(\{P(a_i, b_j)\} \| \{P(a_i)\} \{P(b_j)\})$$

or

$$H(P(x) \| Q(x)) \equiv \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

$$\equiv -H(x) - \sum_x P(x) \log(Q(x))$$

Mutual information:

$$H(A:B) \equiv H(A) + H(B) - H(A, B)$$

• measure of the correlation between A & B

$$H(A:B) = \sum_{i,j} P(a_i, b_j) \log(L(a_i | b_j))$$

$$\sup(H(A), H(B)) \leq H(A, B) \leq H(A) + H(B)$$

Define: $H(B|a_i) \equiv - \sum_j P(b_j | a_i) \log(P(b_j | a_i))$

Info associated w/ B given we know $A = a_i$

$$H(B|A) = - \sum_i P(a_i) H(B|a_i)$$

$$= - \sum_i P(a_i) \sum_j P(b_j | a_i) \log(P(b_j | a_i))$$

$$= - \sum_{i,j} P(a_i, b_j) \log(P(b_j | a_i))$$

$$= \sum_{i,j} P(a_i, b_j) \log(P(a_i, b_j) / P(a_i))$$

$$= \sum_{i,j} P(a_i, b_j) [\log(P(a_i, b_j)) - \log(P(a_i))]$$

$$= \sum_{i,j} P(a_i, b_j) \log(P(a_i, b_j)) + \sum_i P(a_i) \log(P(a_i))$$

$$\therefore H(B|A) = H(A, B) - H(A)$$

Jayne's maximum entropy:

i) Take a die w/ 6-sides $1 \leftrightarrow 6$. $\frac{21}{6}$

$$\text{Mean} = \frac{1}{6} \sum_{i=1}^6 i = \frac{1}{6} (1+2+3+4+5+6) = \frac{21}{6} = 3 + \frac{3}{6} = 3.5$$

• Max entropy fun states:

$$H_e = - \sum p_n \ln(p_n)$$

Subject to the constraints that the prob. sum to unity and mean ≈ 3.47

→ Introduce a Lagrange multiplier for each constraint (μ, λ)

and add: Const. = Value

$$\Rightarrow (\text{Const.} - \text{Value}) = 0$$

$$\tilde{H} = H_e + \lambda (1 - \sum_{n=1}^6 p_n) + \mu (3.47 - \sum_{n=1}^6 n p_n)$$

$$\therefore d\tilde{H} = - \sum_{n=1}^6 \ln(p_n) - \sum_{n=1}^6 1 - \lambda \sum_{n=1}^6 1 - \mu \sum_{n=1}^6 n = 0$$

$$= - \sum_{n=1}^6 [\ln(p_n) + 1 + \lambda + \mu n] = 0$$

$$\therefore \ln(p_n) = -1 - \lambda - \mu n$$

$$p_n = e^{-(1+\lambda)} e^{-\mu n}$$

$$-(1+\lambda) - \mu n$$

$$p_n = e^{-(1+\lambda)n}$$

Now, use $\sum_{n=1}^6 p_n = 1$ to solve for one of these variables (λ):

$$\sum_{n=1}^6 p_n = e^{-(1+\lambda)} \sum_{n=1}^6 e^{-\mu n} = 1$$

$$\therefore e^{1+\lambda} = \sum_{n=1}^6 e^{-\mu n}$$

$$p_n = e^{-\mu n} / \sum_{n=1}^6 e^{-\mu n}$$

Now: $3.47 = \sum_{n=1}^6 n \frac{e^{-\mu n}}{\sum_{m=1}^6 e^{-\mu m}}$

$$5\mu + 4\mu + 3\mu + 2\mu + \mu$$

$$3.47 \sum_{m=1}^6 e^{-\mu m} = \sum_{n=1}^6 n e^{-\mu n}$$

$$3.47 (e^{-\mu} + e^{-2\mu} + \dots + e^{-6\mu}) = (e^{-\mu} + 2e^{-2\mu} + \dots + 6e^{-6\mu})$$

$$3.47 (e^{5\mu} + e^{4\mu} + \dots + 1) = (e^{5\mu} + 2e^{4\mu} + 3e^{3\mu} + 4e^{2\mu} + 5e^{\mu} + 6)$$

$$2.47e^{5\mu} + 1.47e^{4\mu} + .47e^{3\mu} - .47e^{2\mu} - 1.47e^{\mu} - 2.47 = 0$$

$$= 1 + \mu$$

$$\mu \approx .10$$

$$\approx 2.47(1 + 5\mu) + 1.47(1 + 4\mu)$$

Ex: Populations in Finland:

	Swedish	Finnish	
Brown	x	$.1 - x$	b
Blue	$.06 - x$	$.84 + x$	a
	6%	94%	

$$H_e = - \sum_i p_i \log(p_i) = - [x \log(x) + (.1 - x) \log(.1 - x) + (.06 - x) \log(.06 - x) + (.84 + x) \log(.84 + x)]$$

Maximize H :

$$dH=0 \Rightarrow \boxed{X = .006}$$

Interpretation: Without any additional information we should use 6% as the % of Firms who are native Swedish speakers w/ Brown Eyes.

* Relationship to Stat Mech:

i) Λ -configurations (states) of a system w/ associate energies $\{E_n\}$

→ Would like to know $\bar{E}$ or $\langle E \rangle$

• Maximum Entropy method tells us to max.

$$H_e = - \sum_n p_n \ln(p_n)$$

• We know: $\sum_n p_n = 1$ and the mean/average Energy: $\sum_n p_n E_n = \langle E \rangle$

$$\left\{ \begin{array}{l} \text{Ex: } p_n = 1/6 \text{ for a die, so that} \\ \sum_n (1/6) = (1/6)(1 + \dots + 6) \end{array} \right.$$

→ Introduce Lagrange Multipliers:

$$\bar{H} = H_e + \lambda (1 - \sum_n p_n) + \mu (\bar{E} - \sum_n p_n E_n)$$

Vary w.r. to p_n :

$$d\bar{H} = \sum_{i=1}^n \left\{ - \ln(p_i) + 1 - \lambda - \beta E_i \right\} dp_i$$

$$\sum_i \left(\ln(p_i) + 1 + \lambda \right) + \beta \sum_{i=1}^n E_i = 0$$

$\Rightarrow$ Term by term, or we would have to find an orthogonal f^n for $\ln(p_i)$. I don't know what that'd be but it's out there

$$\ln(p_i) = -1 - \lambda - \beta E_n$$

$$\therefore p_n = \frac{e^{-(1+\lambda)}}{e^{-\beta E_n}}$$

• Can eliminate λ via $\sum_n p_n = 1$

$$e^{-(1+\lambda)} \sum_n e^{-\beta E_n} = 1$$

(Partition f^n)



$$\therefore p_n = e^{-\mu E_n} / Z(\beta)$$

(Boltzmann distribution)



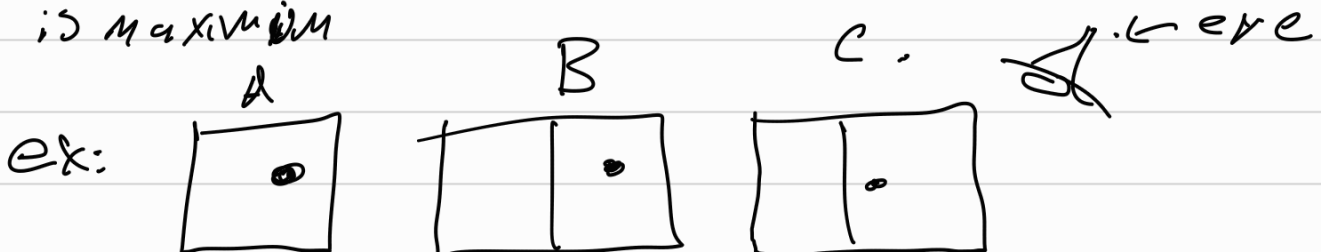
$$Z(\beta) \equiv \sum_n e^{-\beta E_n}$$

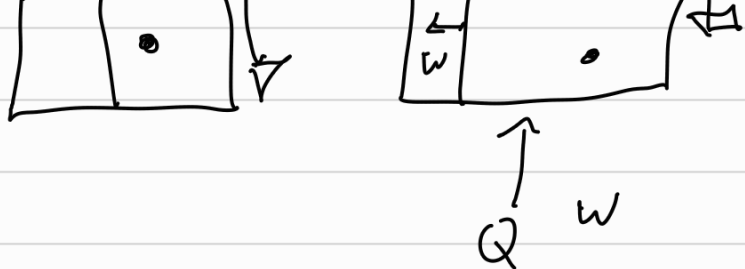
$$\beta \equiv \frac{1}{k_B T}$$

Information + thermo:

* Information is stored as an arrangement of physical systems and these systems are subject to physical laws.

2nd law of thermo: During real physical processes the entropy of an isolated system always increases. In the state of equilibrium, the entropy obtained is maximum





• Volume: V_0 (A), temp T

(B) Heat engine starts and we half the volume
 $V_0 = \frac{1}{2}V_0 + \frac{1}{2}V_0$

(C) Observe which side the molecule is in

(D) Add a weight to the top of the partition

Assume: $PV = nRT \Rightarrow nR = k_B \approx 1.38 \times 10^{-23} \text{ J K}^{-1}$
 $PV = k_B T$

• Equal probs. for the molecule to be in either
 $\rightarrow 0 \text{ or } 1$

$$H = -\frac{1}{2} \ln\left(\frac{1}{2}\right) - \frac{1}{2} \ln\left(\frac{1}{2}\right) = -\ln\left(\frac{1}{2}\right)$$

$$H = \ln(2)$$

The information obtained in this case: non-varying

$$W = \int \vec{K.E.} \cdot d\vec{r} \quad \left\langle \frac{1}{2} m v^2 \right\rangle = \frac{f}{2} k T$$

The molecule has two D.O.F.

$$\therefore \left\langle \frac{1}{2} m v^2 \right\rangle = k T$$

$$W = \int P dV$$

$\rightarrow$ Isothermal expansion

$$W = \int_{V_0/2}^{V_0} P dV$$

i.e. we'd like to see how the molecule "pushes" the wall. We put in heat Q so that the molecule does not "lose" energy.

$$Q = W$$

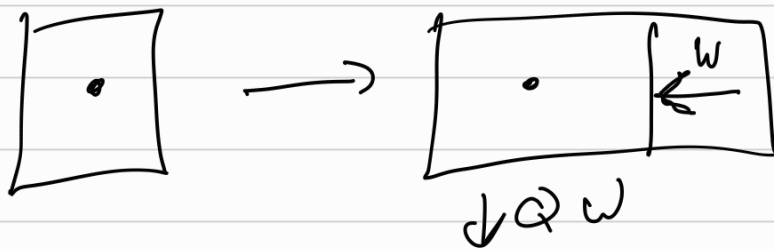
$$\int_{V_0/2}^{V_0} \frac{n}{V} dV = k \ln(V) \Big|_{V_0/2}^{V_0} = kT \ln(2) = W$$

$$\Delta S = \frac{Q}{T} = \frac{kT \ln(2)}{T} = k \ln(2)$$

* Erasing a bit of information requires the dissipation of at least $k_B T \ln(2)$ of energy.

• isothermal compression of a 1-mol. gas

$$W = - \int_{V_0}^{V_0/2} P dV = k_B T \ln(2)$$

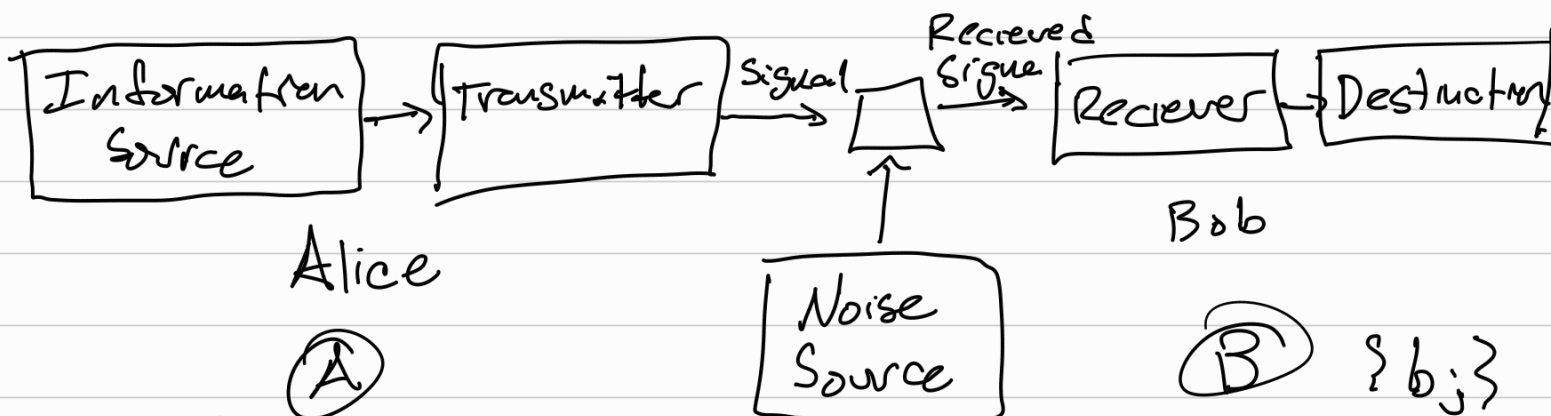


Communication theory:

Alice: transmitter

Bob: Receiver

* Shannon's schematic diagram of a general communication system



$\{a_i\}$: possible choices of messages

$P(a_i)$:

• Communication determined by conditional prob.

$\{P(b_i|a_i)\}$

→ To Bob $\{P(a_i|b_i)\}$ is important: prob. of

Alice sending a_i ; given that he has received b_j .

$$P(a_i, b_j) = \delta_{ij}$$

• Noise will induce errors

Shannon's Noiseless coding thm:

N bits: $01011\dots$
 $\underbrace{\hspace{1.5cm}}_N$

$$P(0) = p$$

$$P(1) = 1-p$$

all mutually independent

• There are 2^N possible arrangements of strings: encode 2^N messages

* Binomial distribution: $n: 0$

$N-n: 1$

$$p^n (1-p)^{N-n}$$

$$P(n) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n}$$

$$= \binom{N}{n} p^n (1-p)^{N-n}$$

• Consider: $n = pN$

$$W = \frac{N!}{n!(N-n)!} \quad (\text{distinct strings})$$

$$p^{pN} (1-p)^{N-pN} = p^{pN} (1-p)^{N(1-p)}$$

$$e^{pN \ln(p)} e^{N(1-p) \ln(1-p)} = \exp [pN \log(p) + N(1-p) \log(1-p)]$$

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

$$\sum_n P(n) = 1 \quad \Bigg| \quad p = \frac{n}{N}$$

Then

$$W = \binom{N}{n} \text{ states.}$$

$$\log(W) \approx N [-p \log(p) - (1-p) \log(1-p)] = N H(p)$$

$$W = 2^{NH(p)}$$

• $NH(p)$ bits to encode a message

→ original message can be compressed by a factor of $H(p)$.

⇒ W/o estimates: $W = 2^{N(H(p) + \delta)}$
 $\lim_{N \rightarrow \infty} \delta = 0$

ex:

	A	B	C	D
$P(i)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$
	00	01	10	11

• Use two bits for each letter:

• A sequence of N bits would require $2N$ bits

Entropy: $H = -\left(\frac{1}{2} \log\left(\frac{1}{2}\right) + \frac{1}{4} \log\left(\frac{1}{4}\right) + \frac{1}{4} \log\left(\frac{1}{4}\right)\right)$

$$H = \frac{1}{2} (1) + \frac{1}{4} (2) + \frac{1}{4} (2) = \frac{7}{4}$$

$$H = 7/4 \text{ bits}$$

H : Quantity of information

Shannon's limit: $\frac{7}{4} N$ bits

→ This limit has

$$A = 0$$

$$B = 10$$

$$C = 110$$

$$D = 111$$

• AVERAGE # of bits:

$$N\left(\frac{1}{2}1 + \frac{1}{4}2 + \frac{1}{8}3 + \frac{1}{8}3\right) = \frac{7}{4}N$$

* (N_0 bits) in message compressed so no redundancy remains.

2^{N_0} can be transmitted w/ prob: $\frac{1}{2^{N_0}}$

• Prob. of bit flipping: q
An average of qN_0 of the Bits received by Bob will be flipped.

* OBSERVER: $\binom{N_0}{qN_0}$

→ Telling Bob where the errors are must be at least $\log(E)$ bits

$$\log(E) \approx N_0 [-q \log(q) - (1-q) \log(1-q)]$$

$$= N_0 H(q)$$

∴ $H(q)$: Entropy associated w/ the prob. for a single bit error.

• Adding continuous observers:

$$N_0 H(q) \quad 1 - \text{obs.}$$

$$N_0 H^2(q) \quad 2 - \text{obs.} \dots$$

$$N_0 H^3(q) \quad 3 - \dots$$

⋮

⋮

$$N_0 \sum_{n=1}^{\infty} H^{(n)}(q)$$

Taylor expansion of $\frac{1}{1-x} = \sum_{n=1}^{\infty} x^n$

∴ Total number of required b.f.s

$$N = N_0 \sum_n H^{(n)} = \frac{N_0}{1-H(q)}$$

• In the large N limit each binary digit can carry no more than

$$\log(2^{N(1-H(q))})/N = 1-H(q)$$

bits of info.

(A)

N $\{P(a_i)\}$
 $\{a_i\}$
 Sends: $2^{NH(A)}$
 likely strings

(B)

$\{b_j\}$
 $P(b_j|a_i)$
 Receives: $2^{NH(B)}$
 likely strings

of messages that can be sent through the channel
 limited by $2^{N[H(B) - H(B|A)]} = 2^{N[H(A) - H(A|B)]} = 2^{NH(A:B)}$

Capacity [of a channel]: $C \equiv \sup(H(A:B))$

/